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TOPIC:

A WEB APPLICATION FOR AGRICULTURAL MARKETING

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DECLARATION

We, **ASANTE RICHMOND** and **SAGOE BRIGHT MARCUS** declare that this submission is our work towards the award of a BSc. in Computer Science and Informatics and that to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the University, except where due acknowledgement has been made in the text.

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DEDICATION

This report is dedicated to the members of our families, especially to our parents who have been our financial source through all our academic pursuits up to this extent, and for the support and sacrifice of our family members during our project work.

ACKNOWLEDGEMENT

First, we would like to thank God Almighty for giving us the strength, knowledge and opportunity to undertake this project and to persevere and complete it satisfactorily. This achievement would not have been possible without his blessings and abundant grace.

We would like to express our special thanks of gratitude to our supervisor Mr Quayson Ebenezer as well as the school authorities and the staff of UENR, especially the Computer Science and Informatics Department who gave us the golden opportunity to do this wonderful project, which has broadened our knowledge. We are thankful to them.

Last but not the least, we would like to thank our family members, program mates and friends who helped us a lot in our educational career and in finalizing this project within the limited time frame.

ABSTRACT

Agricultural marketing incorporates all activities which add value to agricultural products as they move from zones of agrarian generation to extreme consumption points. A successful and effective marketing framework is required to guarantee fair returns to partners but unfortunately, small consideration is paid to this segment in Bibiani Anhwiaso Bekwai (BAB) Municipality. This brings about welfare risk to stakeholders as price variation and value are common phenomena within the municipality. The agricultural marketing strategic vision and effectiveness can be achieved with the help of electronic commerce-related concepts, to ensure successful sales and purchase activities. Due to poor post-harvest management practices, almost 25-40% of produce is wasted which cause considerable value losses. This system will enable farmers to sell their perishable and non-perishable agricultural products online to buyers. It is aimed at providing a wider market and helping them to not restrict themselves to the local market. This report is made up of some experiences, which were made during the development of the online market system. Specific about the project is the fact that it started as an attempt to develop a customizable standard web Application for smallholder farmers in the BAB municipality to showcase their products to buyers. The report details the situation before and after the reform and discusses the experiences made through the ongoing development. These experiences relate to three areas namely; software project management, prototyping, and testing.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the study

Agriculture is the source of livelihood for most people in developing countries, and it generates income and employment for the vast population in these countries. Agriculture provides raw materials for both local and foreign industries. In Ghana, it is the most important economic sector, employing more than half of the population on a formal and informal basis and accounting for almost half of Gross Domestic Product (GDP) and export earnings. Even within the larger economy, the 2010 census estimates that agriculture remains the largest employment sector, offering jobs to about 41.6 per cent of the economically active population older than 15 years (Ghana Statistical Service, 2012, p. 75, as cited in Dzanku & Aidam, 2013). This sector has extraordinary potential to back the national economy in future if due consideration is giving to unravel rising issues. Sustainable agricultural development and improvement depend on how concerns of different partners, especially agriculturists who are confronting a few dangers within the production and marketing of their produce, are adequately addressed (Ghafoor et al., 2010). According to Ghana Labor Force Report (2015), there were 9.3 million people who were; formally employed in 2015. Of the total number, 3.3 million (about 36 per cent) engaged in agriculture. However, in the rural areas, the aggregate employment recorded was 4.6 million (49.1 per cent of total employed). Of the aggregate rural employees, 70.6 per cent were working in the agriculture sector. Agriculture needs effective utilization of technology to quicken the production and employability of people (Radhika, 2018). The share of agriculture in the overall economy reached its peak in the late 1970s and early 1980s when the sector contributed about 60 per cent to total GDP (Breisinger et al., 2015). The agriculture sector is important to the development of Ghana since it is this area that

plays a singular role in leading the growth and structural transformation of the economy of the country by guaranteeing food security, providing crude materials for industry, creating employment and generating foreign exchange earnings, all of which are fundamental for the country's socio-economic development (Akudugu et al., 2013). Marketing activities enable a producer to discover new buyers, build and keep up connections with current buyers, and access market research to oversee supply, anticipate demands and establish prices. Agricultural products range from staple to cash crops; however, there's an emphasis on cash crops marketed by smallholder farmers and provide income to support rural households. In Ghana, larger-scale participants in agricultural activities such as commercial farmers, traders, wholesalers and exporters often have well-developed markets. Smallholders, in any case, are still exceptionally restricted in their marketing capabilities. They are individual households who depend on only one buyer and have little access to price or other demand-related information and little power to negotiate.

The producers of food and other agricultural products in the BAB municipality depend solely on the sale of these crops as their source of income. However, they have no available platform to showcase their products beyond the local market, where only small quantities of their commodities are purchased. This project aims to design and implement a web application that will enable farmers to showcase their products. It will also provide a mechanism for buyers to purchase agricultural products in bulk and small quantities at the comfort of their locations.

1.2 Problem Statement

Agricultural Marketing includes all tasks which add value to agricultural products as they move from areas of agricultural production to ultimate urban areas for consumption. A successful and productive marketing system is required to ensure fair returns to farmers, but unfortunately, little consideration is paid to this sector across Ghana. For instance, customers who are willing to buy

an agricultural product such as foodstuffs, charcoal, and the likes in bulk quantities for resale or use often find it difficult because there is no existing platform to check the availability of what they want anytime. Again, industries that require these products also sometimes find it punishing to get these products. They often move in-between communities at higher costs and risks. Prices of staple foods are still relatively high, mainly due to the high transportation and handling costs resulting from poor rural infrastructure. During harvest, most of the products go to waste because there is no ready market for them. The farmers charge a low amount of money for their products because there is the fear of those products going to waste because they do not have storage facilities. Also, there is no digital platform where they can showcase their products to buyers. The farmers in the BAB municipality may often be exploited by middlemen who usually purchase these products at very affordable prices for resale at relatively high prices. They are also limited to only the local market where there are only a few buyers of their products.

The above situations call for the introduction of a Web Application for Agricultural Marketing for farmers within the area. This system eliminates the middlemen and arranges facilities for the farmers to showcase their products to buyers. This platform will help wholesalers and consumers to get the products at affordable prices. It will also provide a large market base for agricultural products as a way to boost agricultural activities in the municipality to help better the standard of living of the farmers.

1.3 Objective of the Study

General objectives

The main objective of this project is to design and implement a web application for the marketing and ordering of agricultural products online within the BAB municipality to promote agricultural activities in the area.

Specific Objectives

This research specifically aims to:

1. design and implement an online platform for farmers to showcase their products
2. provide an online platform where customers would be able to order and pay for agricultural products
3. develop and implement a real-time mobile payment system

1.4 Significance of the Study

The online Agricultural Marketing system will provide an effective and efficient means to sell or purchase agricultural products. It will also come up with a large market for farm produce within the BAB municipality. It will be beneficial to the following entities:

Farmers within the BAB Municipal Assembly: Almost all farmers in the municipal assembly produce expecting the sales of their products after harvest. However, the unavailability of ready-market makes it difficult for them to sell those products, causing a loss at the end of the season. With the aid of this system, the farmers will be able to showcase their products to trusted buyers from within and outside the locality which would lead to sustainable growth in agricultural activities in the area.

Buyers: Buyers in the case ranges from an individual firm that depends on agricultural products for production to consumers who buy these products in low and bulk quantities. It is often time-consuming and costly to move between locations to get these products, therefore with the advent of the online agriculture marketing system, a variety of agricultural products would be made available to these buyers in small and bulk quantities.

1.5 Limitations

This study has been constrained by time coupled with other academic work thereby limiting the data collection process.

1.6 Scope of the Study

The proposed system is limited to the showcase of agricultural products, providing a digital-based platform for marketing these products to promote agricultural activities to better the standard of living of the rural settlements in the BAB Municipal Assembly in the Western-North Region where agriculture is the main occupation.

1.7 Organization of Work

This work is in five chapters. Following the general introduction in chapter one, this chapter includes the limitations faced while conducting this research. Chapter two explores the literature review relating to the study of the previous secondary data available on this topic. Chapter three focuses on what the agriculture marketing system is all about. This chapter describes the research design and procedures used for the study. Chapter Four is the system development stage which includes system design and structure, testing and evaluation. The summary, conclusions and recommendations are in are discussed in chapter five.

CHAPTER TWO

LITERATURE REVIEW

2.0 Agricultural Marketing Application Overview

The review of the study will consider the benefits of a marketing application to farmers and consumers, issues associated with web applications, the impact of digital marketing on local markets, features of electronic-commerce applications among other things. The literature also assesses the problems farmers face in the sales of their products in the local market without using any form of technology.

2.1 Online Agriculture Marketing Application: Benefits and Related Issues

The adoption of technology has been beneficial to the progress of agriculture in advanced countries which utilizes computer applications in the field of agriculture from land preparation to the sale of agro products. The marketing of agricultural products plays a vital role in the agro-food supply chain in sub-Saharan Africa. One of the primary choices that farmers got to make is the choice of marketing platform for their products. Different market platforms offer different commodity prices that affect farmers' income and welfare (Adu-Donkor et al., 2021). In the digital world, we can revolutionize the agriculture market end to end digitally. ICT helps smallholders to discover numerous buyers who are willing to pay high prices for producers (Dabbara et al., 2019). The digital market contributes to the successful advancement of products and businesses through computerized channels, which essentially increments the targeting of marketing activities, extends the reach of consumers. Also, it saves the budget and helps to adapt advertising to a particular consumer. This area of marketing is referred to as digital marketing (Nezamova & Krasnoyarsk, 2021). Digitalizing of agriculture has promised to be able to provide a solution to a variety of problems faced by the society in the area of food to cater for the increasing population of the world, reducing the

impact of farming on the environment, ensuring the safety of food through traceability (Adu et al., as in Wolfert et al., 2017). Finally, a few studies in sub-Saharan Africa see the utilization of computerized technology as likely to progress agribusiness along the value chain. Studies carried out in Kenya have shown that numerous issues constrain the capacity of smallholder farmers in Africa to have market access, such as the need for data and marketing channels. In any case, the integration of ICT among smallholders has made a difference to resolve these challenges (Okello et al., 2010). Online purchasing has become popular because of the conveniences it brings to the customers to access or purchase products on a digital marketing platform, irrespective of their locations (Aseri, 2021).

One of the main challenges is the trust factor that must be overcome, for buyers to turn to modern ICT-enabled forums as an option for their relationship-based trade. In addition, there are frequently not accepted or understood grading standards for many of the agricultural products that smallholders produce, meaning that buyers often do not know the precise quality of the crop unless they see it physically. Until an adequate level of trust and quality standards within these forums is established, it may be essential for producers or organizations encouraging the use of these services to serve as an intermediary or broker (USAID, 2013).

2.2 Online Web Marketing Application

A web application is a type of application program designed to run on a web server. It provides users with the same services as those computer programmes accessed on personal computers. Web applications refer to applications accessed via a web browser over a network and developed using browser-supported languages such as hypertext markup language (HTML), JavaScript among others. For execution, web applications depend on web browsers and incorporate numerous recognizable applications such as online retail deals, online auctions, and webmail. The adoption of a

Web applications infrastructure can provide vital processes such as the transfer of funds and updates of the pricing information.

2.2 Features of Online Marketing Application

This type of application has the majority of the features that seem like that of the e-commerce applications. To develop an effective and efficient web marketing application, your startup should have the following key features:

2.2.1 Registration Process

The registration feature of the online marketing app usually annoys users if it is too complex. The registration process of your application should not demand so much information from the user.

2.2.2 Fast Check-out

Online marketing is all about convenience so the checkout process should be optimized to accommodate quick and easy transactions for the users.

2.2.3 Sending push notifications

Customers should be notified of new products, let them know when wanted items are back in stock, host flash sales and personalized marketing campaigns amongst many other things with push notifications. This is an effective way to communicate with your customers in a more direct way than email marketing.

2.2.4 Customer Cart

Carts are an essential element and an incredibly popular feature of a web e-commerce app that allows users to save items for later review whiles they view and compare items before buying. It also compliments offline functionality perfectly so users can bookmark items for purchasing later when they have internet access.

2.2.5 Managing Orders and Returns

The main goal of a web marketing app is to make the customer experience as seamless and efficient as possible. Being able to manage their order through the app with things like delivery tracking, the customer will feel more in control and engaged with the buying process.

2.2.6 Detailed Product Information

Providing more detail will help give customers a more informed view of the product before purchasing. This will offer assistance to diminish returns and possibly despondent clients.

2.3 Mode of Operation

Farmers who stay further away from neighbourhood markets are more likely to utilize mobile phones for agricultural marketing to enhance their access to markets. Hence, the use of mobile phones for agricultural marketing may become a preferred, cost-effective and convenient choice to smallholder farmers as restricted to travelling which may require using more money and/or spending more time to access market information (Katengeza et al. 2011).

The operation of the online agriculture marketing application would be categorized based on the activities that are performed by the various user groups of the application. For that matter, the operations that could be carried out by the farmer will be different from that of the buyer, as well as that of the administrator (admin) to ensure maximum satisfaction at each level of the application usage.

2.3.1 Farmer Operation

The farmers would upload the photos of products and the required details about the product as well as their locations. However, it is subject to the admin's approval.

2.3.2 The Customer Operation

A customer is allowed to access the various products on the site to check the price of the products or evaluate alternative products on the site. A cart is also provided where customers can add items to their wish-list for future review

2.3.2 The Admin's Operation

The admin manages the activities of the other actors of the system (the farmer and the customer). In this case, the admin will view the uploaded products to approve the product or reject it.

2.4.0 Existing Online Marketing Applications

To ensure that efficient application is developed, the features of some existing digital marketing systems currently in use in Ghana and other countries were reviewed. The shortcomings of those systems would be addressed in the proposed system.

Currently, there is no existing computerized marketing application for the smallholder farmers in the municipality to offer their products for sale. The farmers usually go to the nearest market and offer their products to a specific agent. He sometimes asks the farmer to visit the market after some time to collect the cash earned from the sold products. The agent later sells the products to another dealer at a higher price, usually in the same market. All the agents try to make a profit out of the transaction while they might bear no transportation cost, and these farmers would not know the actual price their products are sold by the agents. There is no transparency and no facility is present for the farmers to know the product rates at different markets where they can sell their products for higher profits. On the other hand, farmers have a lot of worries when the products involved are perishable such as vegetable which needs a ready market immediately after harvest. They are usually poised to sell these products at very low prices to avoid loss due to the unavailability of a

digital platform where they can showcase their products to attract buyers from within and outside the municipality.

Existing Agricultural Marketing platforms

Following are some existing agricultural marketing platforms in use in Ghana and other countries which we found very useful to review to close the common gaps during the implementation of the proposed system.

2.4.1 AgroMart by AgroCenta Ltd.

Agromart is a web-based agricultural marketing platform that provides the opportunity for people to become distributors, these people play a key role in the agricultural value chain where they will help the smallholder farmer found in the remotest part of Ghana sell their produce at reasonable market prices, bypassing agents who exploit them. It is a complete agricultural marketing system that was designed to work on the webserver. However, we realized few shortcomings in the Agromart website per the review during the signup and login processes. We realized that both processes do not work even if the user enters valid detail or not. In addition to the gap is the number of products that is available for customers: only three agro products (maize, millet and soybean) are made available for customers to purchase or for farmers to upload which means that the site will not be beneficial to farmers who do produce not any of the mentioned products

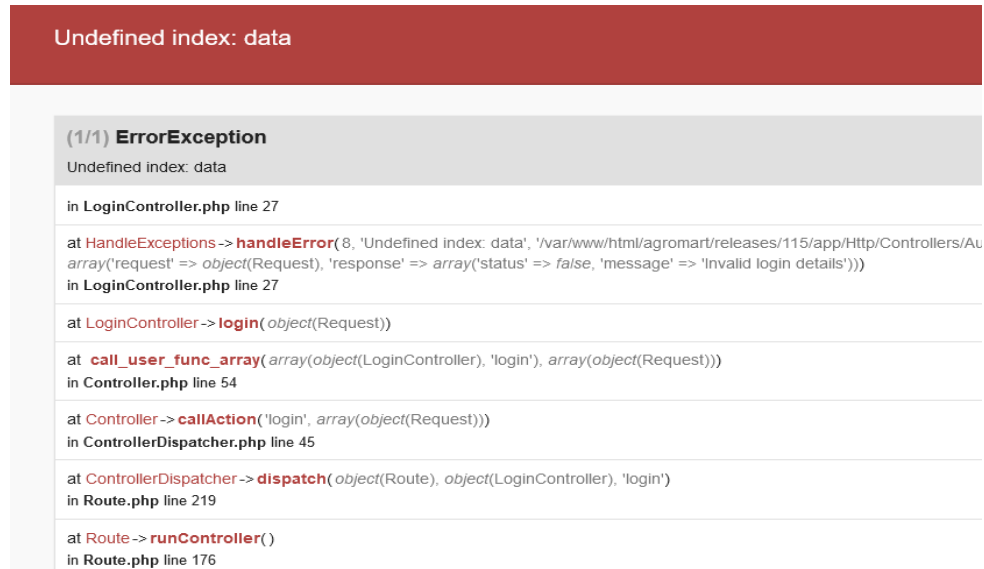


Figure 2. 1: Login interface of AgroMart

2.4.2 Completefarmer

Completefarmer gives cutting-edge technological farming protocols and advancements with an interesting business model and logistics that is revolutionizing farming and creating an end-to-end digital marketplace that enables global industries to source agribusiness commodities, developed to their details, by clients from all over the world utilizing IoT, enormous information and block-chain technology. The gap found in this website is that it does not open on smartphones. It works only on Personal Computers (PCs) which means the user has to get a PC to undertake successful sales or purchase activities.

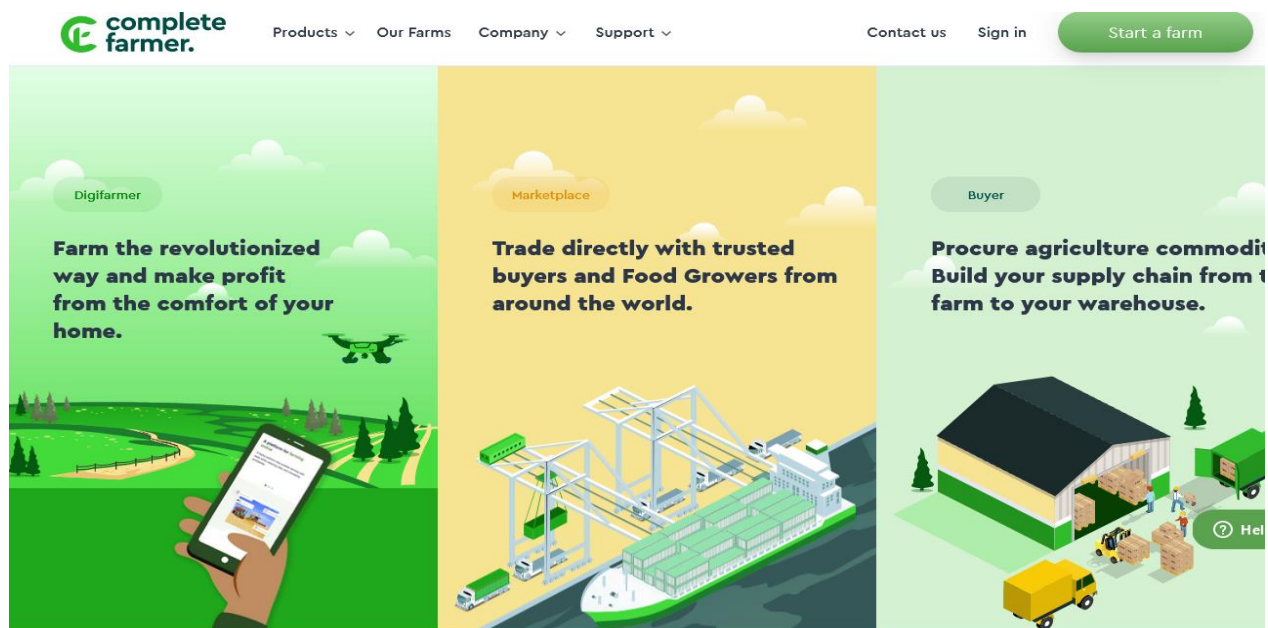


Figure 2. 2: Homepage of Completefarmer

2.5.0 The proposed system

After studying the previous system, we found few gaps in those systems. For instance, to use some of these platforms, it is required to have a good storage facility. However, with the advent of the online agricultural marketing application, these gaps would be closed to ensure high farmer and customer satisfaction. The requirements of the proposed system would match the desire of the various stakeholders of the application (farmers and buyers).

2.5.1 The Admin

The administrator will coordinate activities carried out by the various stakeholders of the system. The admin will be the central point of the application where questions, ratings and suggestions, orders, etc., will be received. In addition to the functions of the admin is adding, managing customers and farmers using the application.

2.5.2 The Farmer

The farmers are provided with a facility to create and manage their accounts, upload their products with detailed descriptions and prices, and manage their posts.

2.5.3 The Customer

The customer has no option to buy the products directly from the farmer. The application will serve as the intermediary between the farmer and the customer. A facility is also provided for the customer to add products to their cart for review or future purchase.

CHAPTER THREE

METHODOLOGIES AND REQUIREMENT ANALYSIS

3.0 Introduction

Chapter three of the project presents a detailed plan of methods for collecting, organizing, analyzing and interpreting data and the techniques for implementing the system so that the objectives of this study are achieved in the end. The chapter deals with the methodology of the study under the following subheadings study area, the population of the study area, sample size, sources of data, methods of data collection, sampling techniques, data analysis and profile of study area.

3.1 Adopted Methodology

Rapid Application Development (RAD) is the methodology used for this project. It refers to a sort of software development technique that uses minimal planning in support of quick prototyping. A prototype is a working model that is practically comparable to a component of the product under development. User modules are created parallel as prototypes and integrated to form the complete product for quicker system delivery in the RAD model. Since there is no nitty-gritty preplanning, it is less demanding to consolidate the changes within the development process. RAD projects follow the iterative and incremental model and have small teams comprising developers, domain experts, customer representatives and other IT resources working progressively on their component or prototype. The foremost vital viewpoint for this model to be effective is to create beyond any doubt that the models developed are reusable. The various stages, namely; analysis, design, build and test are distributed into a series of brief, iterative development cycles within the RAD model. Following are the various phases of the RAD Model: Business Modeling, Data Modeling, Process Modeling, Application Generation, and Testing and Turnover.

3.1.1 Business Modeling

The business model for the product under development is outlined in terms of the flow of data and the dissemination of information between different business channels. Total business analysis is performed to discover the crucial data for business, how it can be obtained, how and when the data is processed, and what are the factors driving the successful flow of information.

3.1.2 Data Modeling

In this phase, the knowledge obtained within the Business Modeling phase is refined into sets of knowledge objects that are crucial for the business. The properties of all data sets are distinguished and characterized. The connection between these data objects is built up and characterized in detail pertinence to the business model.

3.1.3 Process Modeling

The data object sets defined within the Information Modeling phase are changed over to set up the business data flow required to realize particular business objectives as per the business model. The process model for any changes or upgrades to the data object sets is characterized in this stage. Process descriptions for including, deleting, recovering or altering a data object are given.

3.1.4. Application Generation

This model is where all the information gathered is coded, and the system is built using automation tools to change processes and data models into actual models.

3.1.5 Testing and Turnover

This allows for reduced time in the overall testing of the prototypes created. Each model is tested separately to identify and adapt the components quickly to create a good product. It reduces the risk of any paramount issues since most of the programming components have already been tested. The following illustration portrays the RAD Model in detail.

Each of these stages applies some techniques and a sequence of examinations. They include conventions and methods for recording and translating the data with the assistance of diagrams and text.

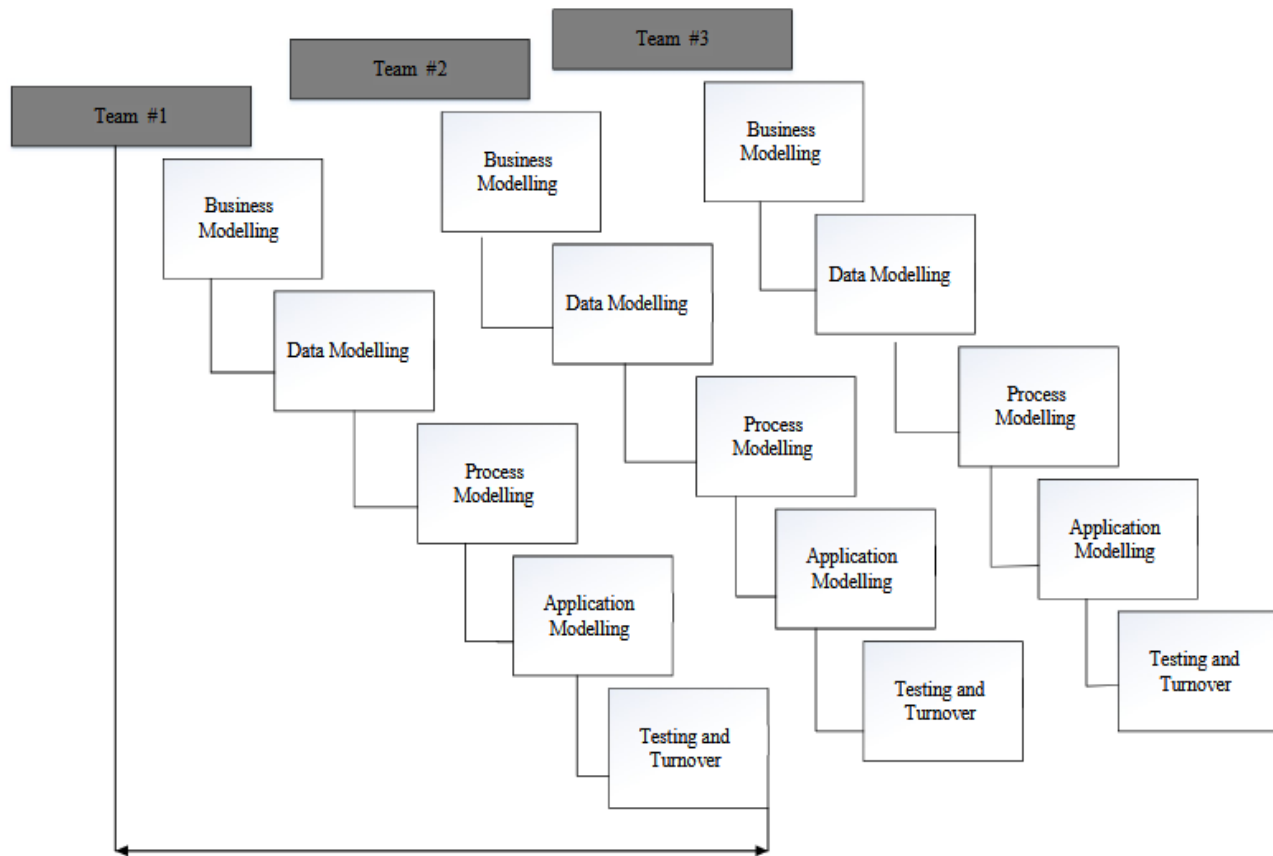


Figure 3. 1: The RAD Models

3.2 Study Area

This project was implemented for smallholder farmers in the BAB Municipality located in the Western-North Region of Ghana. BAB was chosen because the researchers realized that the traditional marketing and agricultural information systems proved inefficient for the farmers to ensure higher production.

3.3 Analysis of Requirements

The application provides an online marketing platform for farmers and consumers to conduct sales and purchases of agricultural products. When developing an effective marketing system, user-level requirements such as user satisfaction, system reliability, system performance, supportability, adoption and integration of the system into a business' workflows, packaging of the system and legal requirements are considered. Three basic functional requirements were identified that would help in the development of the proposed system. These are Inputs, Process and Outputs

3.3.1 Inputs

- Farmer information and product details
- Customer Information
- Payment Information
- Transaction Details

3.3.2 Processes

The farmer uploads products with detailed descriptions and prices. The system records this information with basic customer information such as name, location and contact number. The admin then views the uploaded product and either approve or reject it. After purchasing a product, the customer's transaction details are recorded in the database.

3.3.3 Outputs

- Upload approval or rejection alert via email and short message service (SMS)
- Payment information
- Transaction summary

3.4 Data Collection Technique

In any system analysis, information must be gathered. It can be achieved through any of the following methods: questionnaire, oral interview, reading of textbooks and published articles and Journals, watching television and also surfing the Internet. The data collection technique that was used to source relevant data for analysis, presentation and implementation of the system for this work was a structured interview. A structured interview was conducted to find some specific answers from reliable respondents and key informants such as farmers, middlemen and consumers within the area. The questions and responses obtained from this exercise are shown in **Appendix II**.

3.5 Data Analysis

During the interview stage, much information was gathered concerning the production and marketing of agricultural products, the extent of the local marketing activities of the major stakeholders. The analysis method used was a close inspection of the responses given by the stakeholders.

3.6 Design Phase

The design phase is divided into two levels; the front-end and the back end. The front-end is a graphical interface developed using HTML, CSS, Bootstrap library, JavaScript and the backend which is the database was developed with PHP and MySQL. The logic behind business operations was also implemented in the backend.

3.7 System Architecture

The architecture of a system is the structure of the system, which comprises computer program components, the obvious properties of those components, and the connections among them (Hamdan et al., 2010). System Architecting is a means to make proficient and successful frameworks, by providing an outline, guarding consistency and integrity, and balancing (Muller, 2021).

The System Architect makes a difference for the development group to discover its way in a preferable complex, energetic and questionable world. It includes distinguishing major system components and their communications. The architecture consists of the administrator, farmers, customers, the system database and the application server. The admin, farmer and customer request information from the server, and the server fetches data from the database and provide a response on the requested data to either of them who makes the request. The architectural design of the system is shown below;

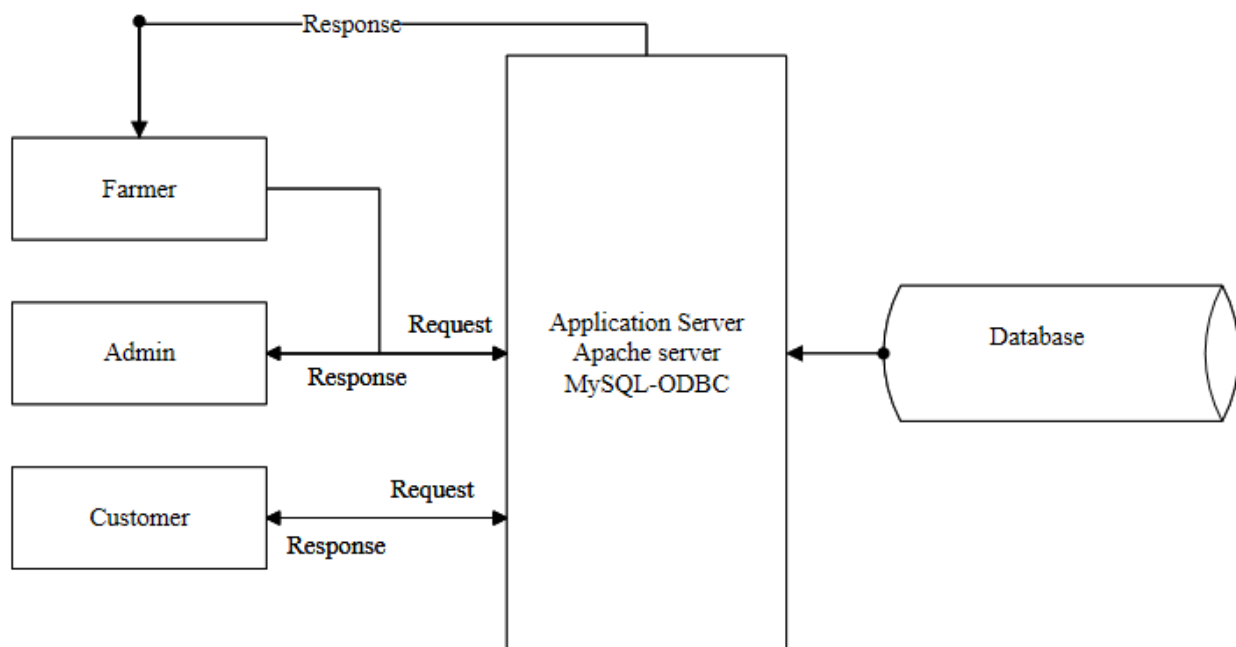


Figure 3. 2: Architecture Design

3.8 Analysis of the System Requirements

Analysis of requirements can be achieved by using a Use – Case diagram. A use case model will be used to describe the requirements of the system in a manner that can be understood by people who are both problem-oriented and solution-oriented (developers and testers for example).

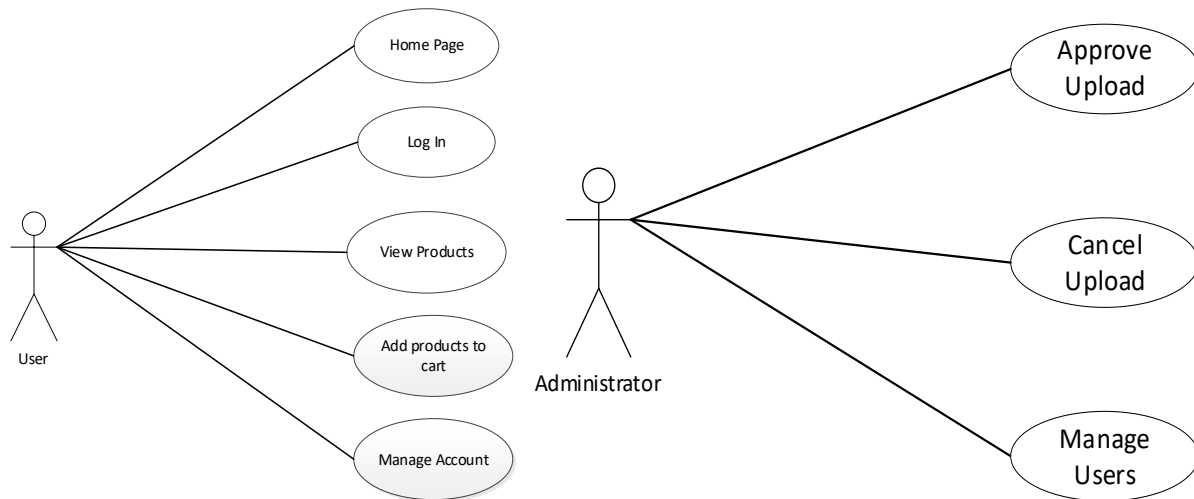


Figure 3. 3: Analyzing the system requirements with a use-case diagram

3.9 Tools Used in Developing the System

The tools and platforms that were used for the development of this application are discussed below.

- Sublime Text:** It is an integrated development environment (IDE). It was used as the primary tool for writing the source codes of the system.
 Sublime Text was the source code editor that was used to code in the various languages
- HTML, Cascade Styling Sheet (CSS):** HTML was used to create the primary contents of the webpage, giving it structure, whiles CSS was also used for background colour, styling, layout, borders, and the essential design of the Graphical User Interface (GUI) of the system.
- JavaScript:** It was used as the Client-side scripting language and also for executing queries and adding special effects to the system
- PHP:** It was the programming language for the backend of the system, which was used for managing databases, session tracking and dynamic contents of the system.
- MySQL:** It was used to create the database and related tables in the database

- **Microsoft Visio:** Microsoft Visio was the tool that was used to draw the various diagrams of the system.

CHAPTER FOUR

DESIGN AND DEVELOPMENT

4.0 Introduction

Software design and implementation is the software engineering process stage whereby an executable software system is developed. For some simple systems, software design and implementation is the software engineering, and all other activities are merged with this process. However, for large systems, software design and implementation are only one of a set of processes (requirements engineering, verification and validation, etc.) involved in software engineering. The creative activity within which software components and their relationships, based on a customer's prerequisites are identified is thought of as Software design. The method of realization of the design as a program is known as implementation (Sommerville, 2011).

Design is the first step in the development of an engineered product or system. It is defined as the process of applying various techniques, skills and principles for the study. It includes the implementation of the theory and the information gathered into a physical solution to solve the real-world situation. The detailed design of the system selected in the study takes place in the design phase. System design is expressed by reviewing the study phase activities and making choices as to which functions are to be performed by hardware, computer program or manually. This is the process through which the initial requirements specified in the study phase are translated into a representation of the software. Initially, the representation depicts analyzing the whole system software. Successive processing leads to a design representation, which is very close to that of the source code. The design stage suggestions are presented as reports to the client.

The following steps were followed in designing the software:

- The problem statement was identified.

- The statements of functions desired from software to be developed were recognized.
- The database design steps were applied to identify the various entities and relationships between them.
- General access methods were then applied to the conceptual schema to get the logical model, which was then further transformed into relational tables.
- The concept of online transactions was then applied to get the logical architecture of the system comprising of different modules each containing all the related functions.

The Symbols Used in the System Dataflow Diagram

The table below shows the various symbols used in the system data flow diagram. It depicts the system, entities within the system, the dataflow and the processes involved and the database of the whole system.

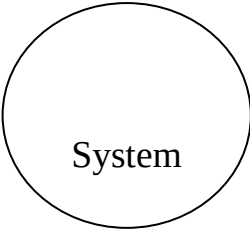
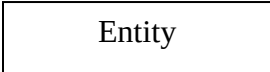
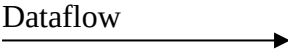
	The system holds the transformation processes that go on, and the manipulation of a data flow within a system is shown using a circle.
	Entities supply input data into the system or utilize its output. However, they may be outside of the system. The entities are represented by a rectangle.
	A data flow shows the flow of information from source to destination. It is represented by a line with arrowheads indicating the direction of the flow.
	The process runs all the activities in the system requesting, responding, activation, etc. and is represented by the process diagram.
	A database holds information within the system. It is represented with a database diagram

Table 4. 1: Symbols used in the system dataflow diagram

4.1 Implementation of the Project

Software implementation is both a process and a model. The implementation process is a sequence of steps that enables the designer to build all aspects of the system. Fundamental implementation standards help the software engineer to explore the design process. There are some principles of software design:

- The design should minimize the intellectual distance between the software and the problem that exists in the real world.
- The design should exhibit uniformity and integration
- The design should be structured to accommodate change

The project was implemented using PHP programming language and MySQL database as the backend. The front end was implemented using HTML, Bootstrap and JavaScript. The various modules of the system implementation are represented in the diagrams below.

4.1.0 System Data Flow Diagrams (DFD)

The data flow diagram is used to depict the flow of information within the system. The system and its users act as a link with all the information flowing into them. Client act as the source since all the request are generated by him. The record is held in the storage spaces depicted by a partially open rectangle. Arrows within the DFD is quite important as it records in a pictorial form all the information flow within the system. We can identify the current information structure of the system with the aid of DFD. The structure of the system can thus be analyzed, and it helps in making improvements within the current system.

4.1.1 Context Flow Diagram

The highest-level view of a system that is analyzed, showing the system as a whole and its inputs and outputs is its System Context Diagram. It shows the interactions between a system and other

actors with which the system is designed to face. It is designed to be an at-a-glance view, showing the system as a single high-level process, with its relationships and external entities.

The agriculture marketing system has three main entities, the customer, the farmer and the admin.

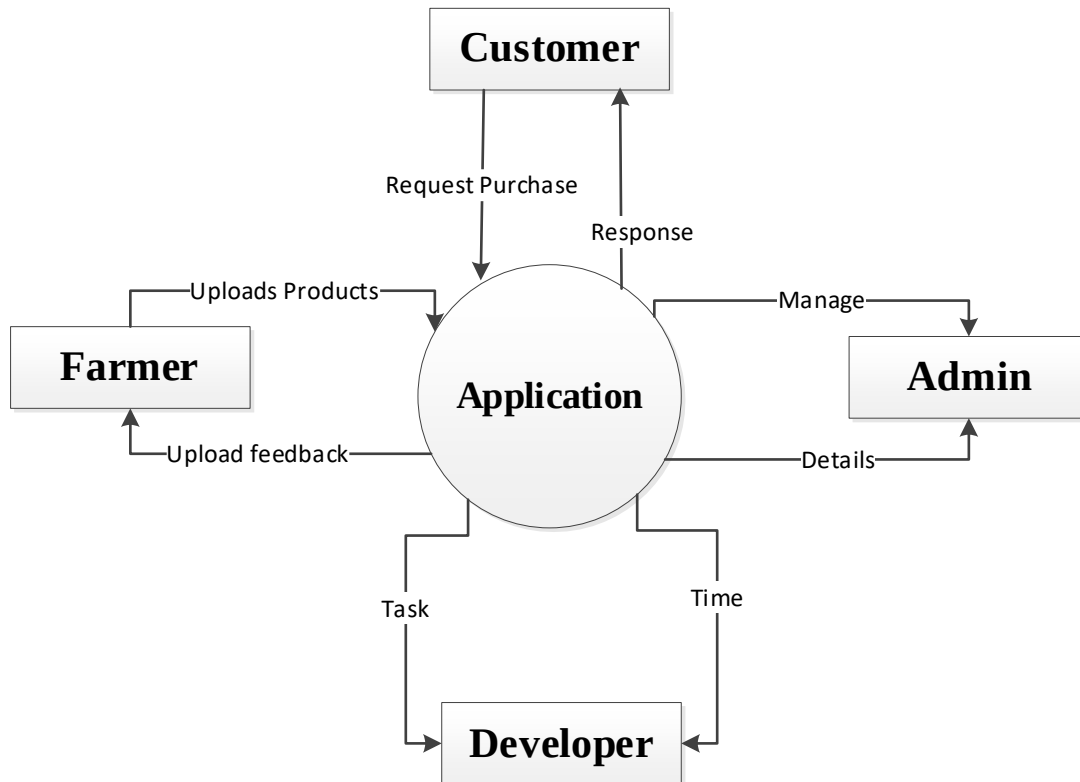


Figure 4. 1: Context Diagram of the System

4.1.2 Customer DFD

The customer DFD depicts the various processes carried out by the customer. It shows the flow of information on the customer's side. He must be authenticated before getting full access to the system. He has to enter his email and password. The system then checks the database to verify whether there is such a user. If the email and password tally with what is in the database, then the customer is given full access to the system.

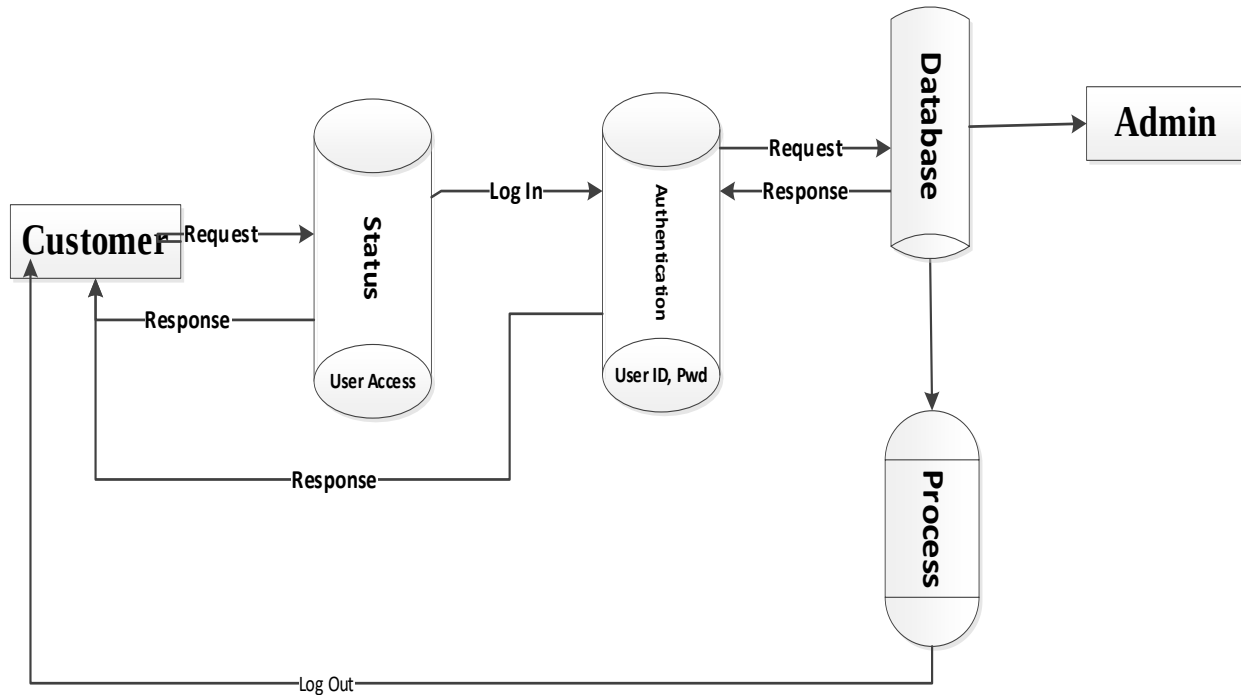


Figure 4. 2: Data flow diagram of the operations performed by the customer

4.1.3 Farmer DFD

The farmer DFD depicts the various processes carried out by the farmer. It shows the flow of information. The farmer gets authenticated before getting full access to the system by providing valid login credentials. The system then checks the database to verify if the farmer exists. When the username and password tally the database, the farmer is giving full access to the system.

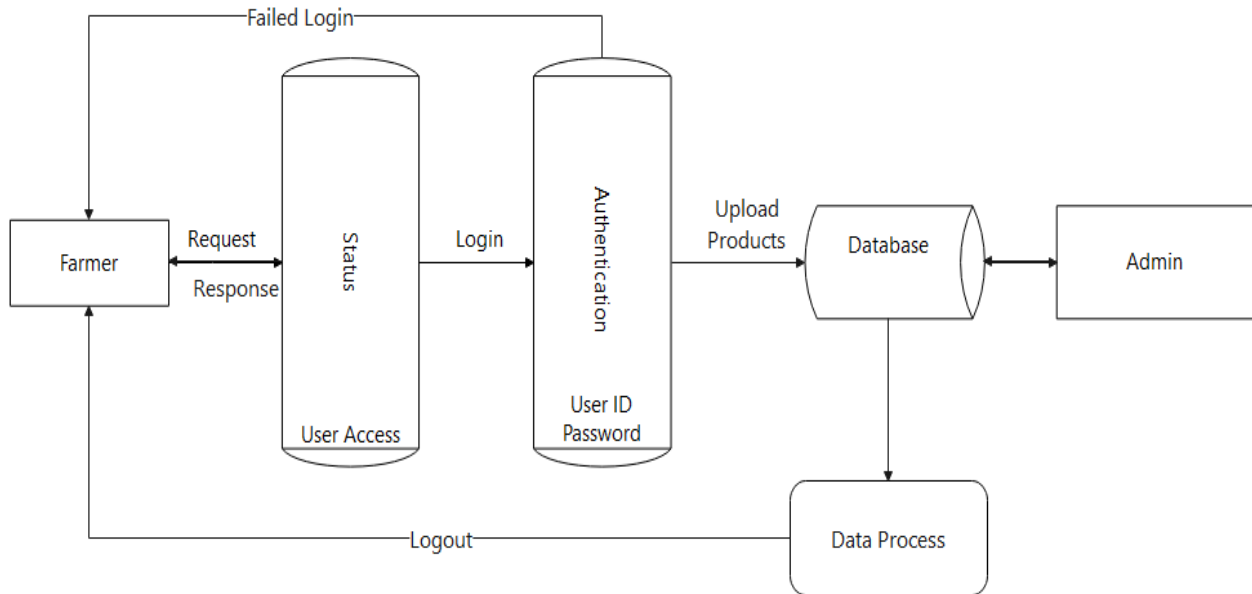


Figure 4. 3: Farmer DFD

4.1.4 Admin DFD

The admin DFD depicts the various processes carried out by the administrator. It shows the flow of information from admin application access to logout. The admin gets authenticated before getting access to the system. The application checks the database to verify the admin's login information and then grant him access to the system.

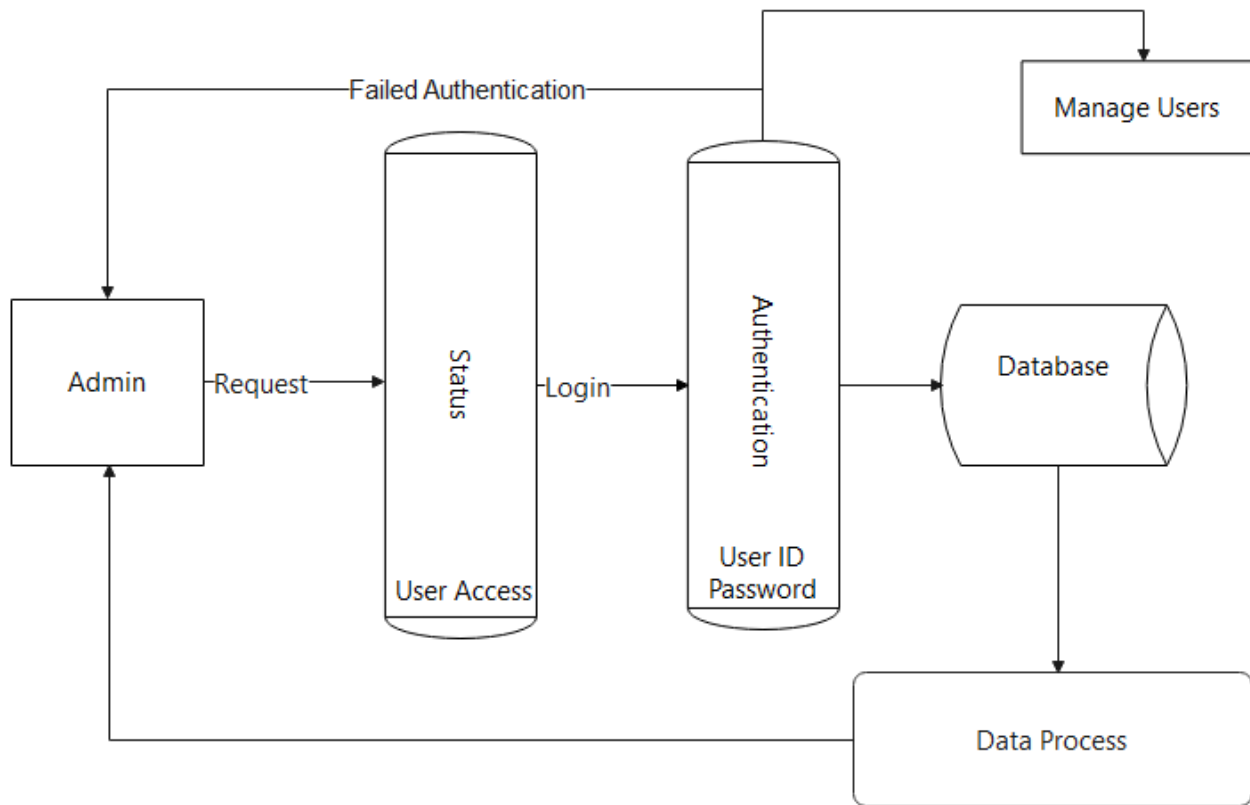


Figure 4. 4: Admin DFD

4.1.5 Analysis of the System Requirements

Analysis of Requirements can be achieved using a Use – Case diagram. A use case model will be used to describe the requirements of the system in a manner in which people who are both problem-oriented and solution-oriented (developers and testers, for example) will understand.

4.1.5.1 Use-Case Diagram

Use cases are a requirements discovery technique first introduced in the object-oriented method (Jacobson et al., 1993). They have now become a fundamental feature of the unified modelling language. In short, a use case recognizes the actors included in the interaction and names the sort of interaction. It can be at that point supplemented by extra information depicting the interaction with the system. The online marketing platform has three actors, the administrator, farmer and the customer. The use case for the various actors is, depicted below.

4.1.5.1.1 Farmer Use Case

The farmer visits the website, and if he does not have an account with the marketing platform, he registers. Otherwise, he logs into the system with his email and password. From thence, the farmer can now upload products to the system, edit or delete his uploads.

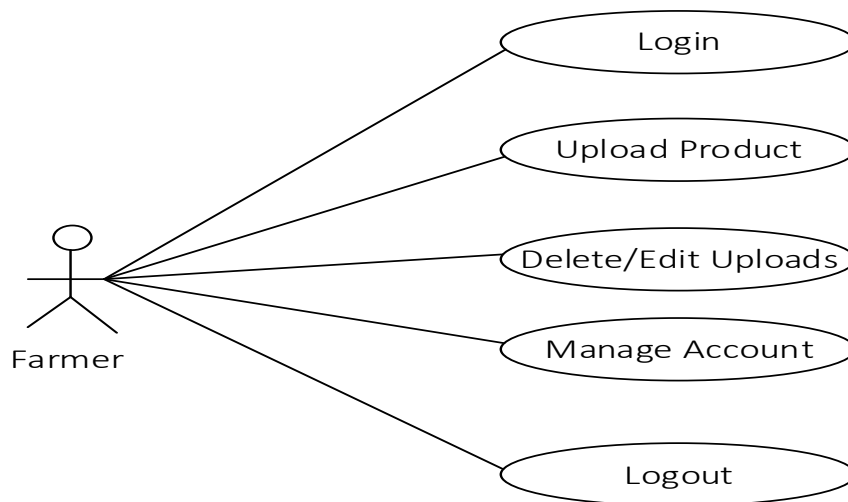


Figure 4. 5: Farmer Use-Case

4.1.5.1.2 Admin Use Case

The admin visits the admin page by first being authenticated. He is permitted to manage the other actors of the system. He can add or delete product categories, approve or reject uploaded products, make payments, manage customer orders and transactions.

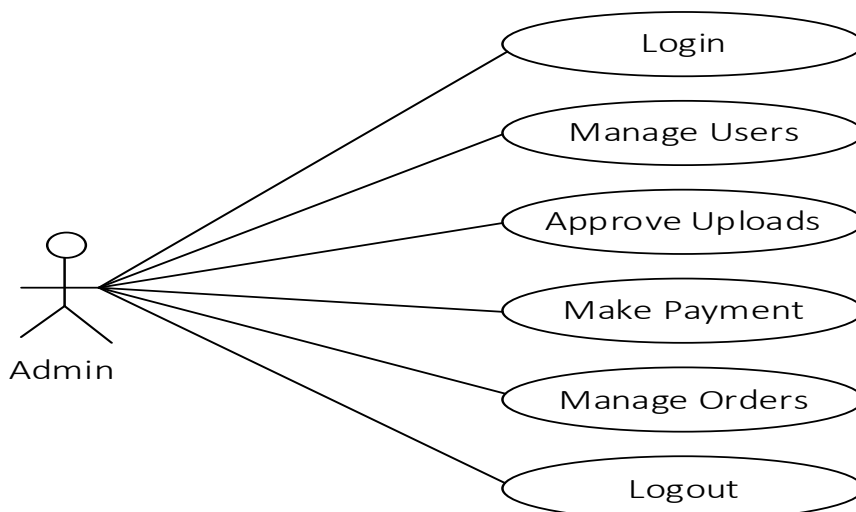


Figure 4. 6: Admin Use-Case

4.1.5.3 Customer Use Case

The customer visits the website, and if he does not have an account with the marketing platform, he registers. Otherwise, he logs into the system with his email and password. From there, the customer can now add products to their cart, purchase products, make payments and manage their account.

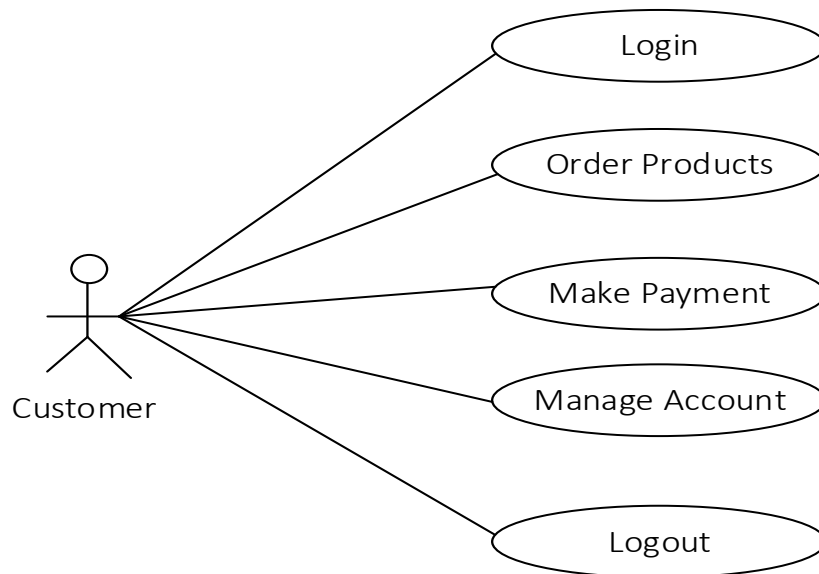


Figure 4. 7: Customer Use-Case

4.2 System Data Flowchart

The flowchart shows the procedures the various actors follow to use the system. Systems flowcharts graphically illustrate the processes, inputs and outputs of a system and are used for the physical modelling of the system.

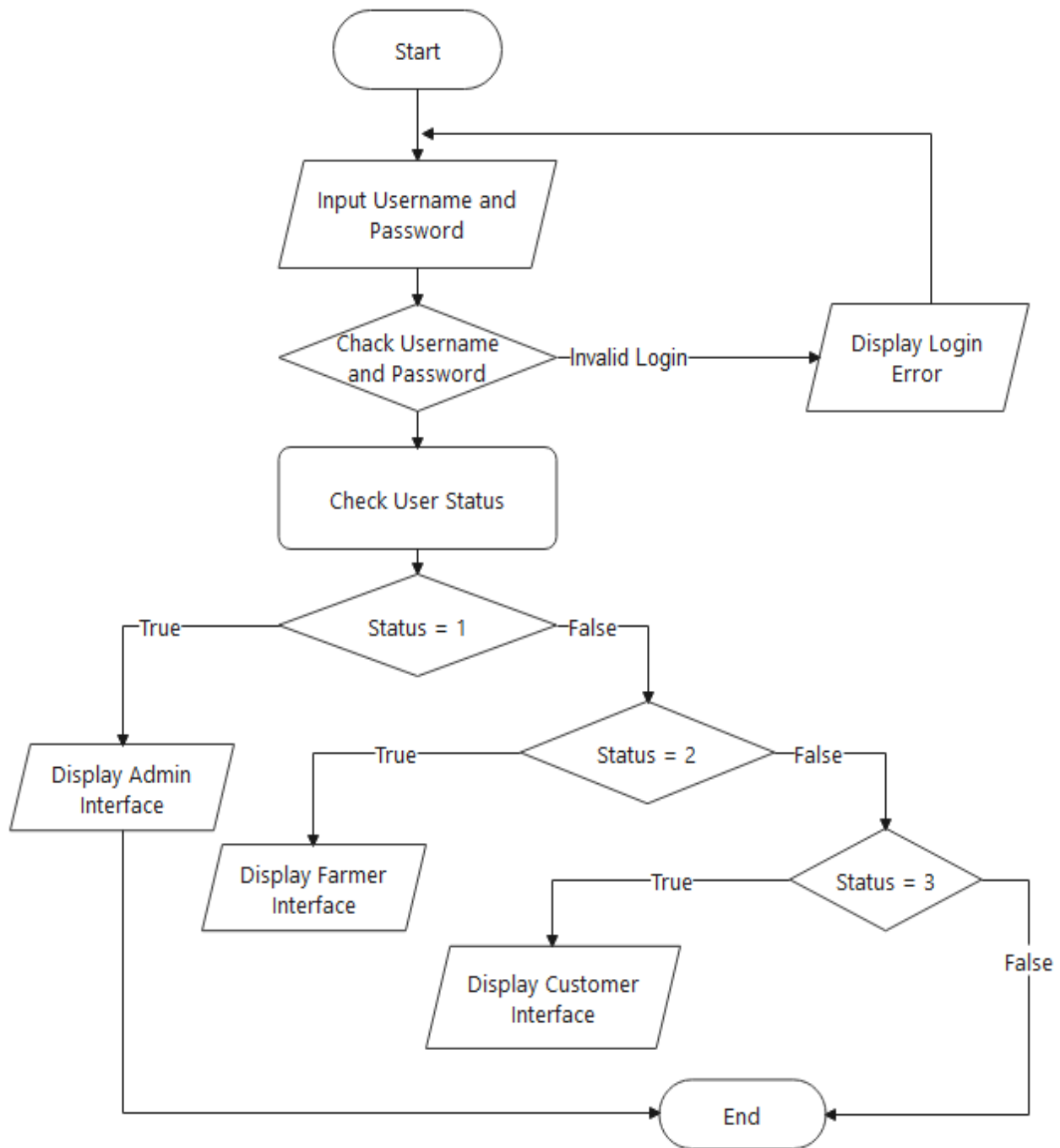


Figure 4. 8: Flow chart of the system

4.3.0 Entity-Relationship (ER) Diagram of the System

ER diagram is a database modelling tool; that allows one to determine the detailed data storage and document all the properties of the designed system. The figure below shows the ER diagram of the system. It includes three main elements of ER diagrams: entities, relationships and attributes.

The entities are labelled admin, users, product and sales. There are various relationships between the entities, such as one-to-one, one-to-many and many-to-many or many-to-one. The attributes are the properties associated with each entity.

Entity Relationship Diagram of the System

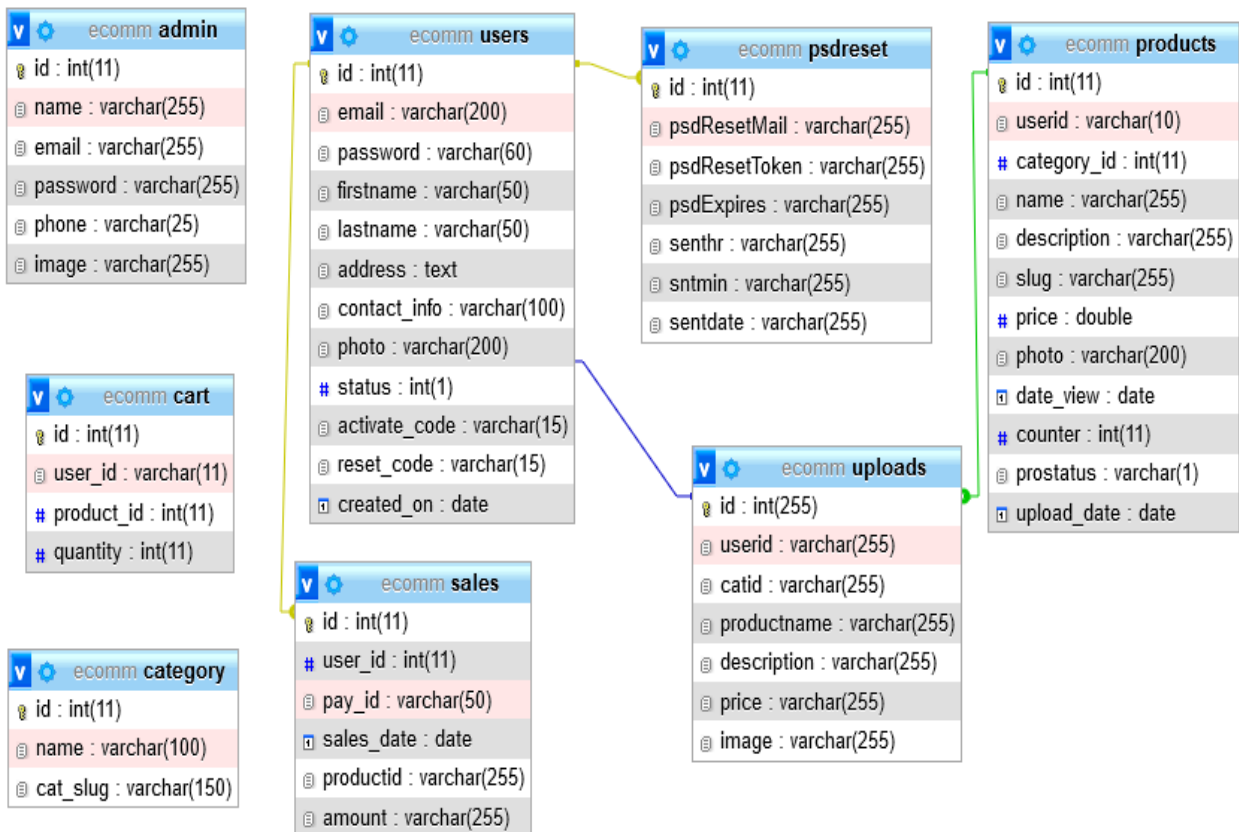


Figure 4. 9: Entity Relationship Diagram of the System

4.3.1 Database Design

The database was designed and implemented using MySQL (Structured Query Language) database. The name of the database is ecom. It consists of ten tables: namely; admin, cart, category, details, products, psdReset, sales, uploads, transactions and users. Each of the tables and forms have their corresponding user interface that enables the user to supply data to the system. The various table design and their attributes are giving in the figure below.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	name	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	email	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 4	password	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
☐ 5	phone	varchar(25)	utf8mb4_general_ci		No	None			Change Drop More
☐ 6	image	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More

Figure 4. 10: admin table

#	Name	Type	Collation	Attributes	Null	Default	Comments
☐ 1	id	int(11)			No	None	
☐ 2	userid	varchar(10)	latin1_swedish_ci		Yes	NULL	
☐ 3	category_id	int(11)			Yes	NULL	
☐ 4	name	varchar(255)	latin1_swedish_ci		Yes	NULL	
☐ 5	description	varchar(255)	latin1_swedish_ci		Yes	NULL	
☐ 6	slug	varchar(255)	latin1_swedish_ci		Yes	NULL	
☐ 7	price	double			Yes	NULL	
☐ 8	photo	varchar(200)	latin1_swedish_ci		Yes	NULL	
☐ 9	date_view	date			Yes	NULL	
☐ 10	counter	int(11)			Yes	NULL	
☐ 11	prostatus	varchar(1)	latin1_swedish_ci		No	0	for checking if the peoduct is enabled or disabled: default value =0
☐ 12	upload_date	date			No	current_timestamp()	

Figure 4. 11: product table

☐	1	id 	int(11)	No	None
☐	2	email	varchar(200) latin1_swedish_ci	No	None
☐	3	password	varchar(60) latin1_swedish_ci	No	None
☐	4	firstname	varchar(50) latin1_swedish_ci	No	None
☐	5	lastname	varchar(50) latin1_swedish_ci	No	None
☐	6	address	text latin1_swedish_ci	No	None
☐	7	contact_info	varchar(100) latin1_swedish_ci	No	None
☐	8	photo	varchar(200) latin1_swedish_ci	No	None
☐	9	status	int(1)	No	None
☐	10	activate_code	varchar(15) latin1_swedish_ci	No	None
☐	11	reset_code	varchar(15) latin1_swedish_ci	No	None
☐	12	created_on	date	No	current_timestamp()

Figure 4. 12: users table

4.3.2 System Security Implementation

The proposed system is secured such that one can only get access to the system with email and password that is only exclusive to users. Data security between the user's browser and the system web server is handled through a security protocol called Secure Sockets Layer (SSL). SSL gives data encryption, server confirmation, and message integrity for a Web connection. Security concerns have been addressed from every angle within the architecture of the system. Implementing SSL security protocol on the Web server and client browser ensures that authenticated data has been received from the client. For authentication, an encryption technique called password_hash is used to save user credentials in the database during registration. It converts the user password into lengthy random characters before sending it to the database. However, if a user wants to access

the system, the password he enters is compared to what is in the database using a technique called password_verify. The user is allowed access to the system if both passwords are the same.

4.4 Testing and Evaluation of the Project

The test plan used in this project was to test the modules, the subsystems, and after that, the complete system. It was in two parts: unit and integration and final testing.

4.4.1 Test Data, Expected versus Actual Test Result

In the subsystem level test, each subsystem is taking as the test data. Whereas at module level test, each module is taking as the test data. The complete system is also taking as the test data amid the system testing. During the testing, the anticipated results were realized and compared to the actual results that came up. These step-by-step processes were done in all the stages of the testing processes and are in this manner presented in the following tables.

4.4.2 Testing the database

The database was tested by providing sample data to each of the database tables and then confirming that the supplied data reflected in the database

Signup page testing

Test Data	Expected Results	Actual Results
Signup form	It was expected to see the form displayed and ready to accept input from the user.	The signup page was displayed, and supplied data reflected in the database

Table 4. 2: Testing the Signup Page of the system.

Login Page Testing

Test Data	Expected Results	Actual Results

Login Form	The expectation was to see the login form displayed and ready to accept input from the user.	The page displays and the supplied data reflected in the database
------------	--	---

Table 4. 3: Testing the login page of the system.

Product upload page testing

Test Data	Expected Results	Actual Results
Product upload form	It was expected to see the form displayed and ready to accept input from the user.	The form displays, and the supplied data reflected in the database

Table 4. 4: Testing the Product Upload Page of the System.

Product approve page Testing

Test Data	Expected Results	Actual Results
Product approve form	The expectation was to see the form display with the products pending approval.	The approval page displayed, and the pending upload in the database reflected as well

Table 4. 5: Testing the Signup Page of the system.

User cart page testing

Test Data	Expected Results	Actual Results
------------------	-------------------------	-----------------------

User cart form	The expectation was to see the form displayed with the products in the customer cart in the database.	The cart page displayed all user added products in the database.
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Table 4. 6: Testing the Sales of the system

Product Category page testing

Test Data	Expected Results	Actual Results
Product Category Form	It was expected to see the form displayed and ready to accept input from the user.	The product category page was displayed. the supplied data reflected in the database

Table 4. 7: Testing the Product Category of the system.

4.4.3 Integrated Testing

All the modules in each subsystem were integrated into one subsystem and tested after the unit and general testing. All the subsystems were integrated into one system and similarly tested. The test results were positive, as the whole system satisfied its objectives.

4.4.4 System Evaluation

The system performance was satisfactory, and it is simple to use. Performance evaluation was carried out in all the testing processes, including the unit and general testing. After carrying out the performance evaluation, the system proved to be efficient and reliable.

4.5 Sample of Output

Homepage of the website

It is the first page that displays if a user types the web address in a browser. It does not require any authentication to access this page. It shows approved products to the customers.

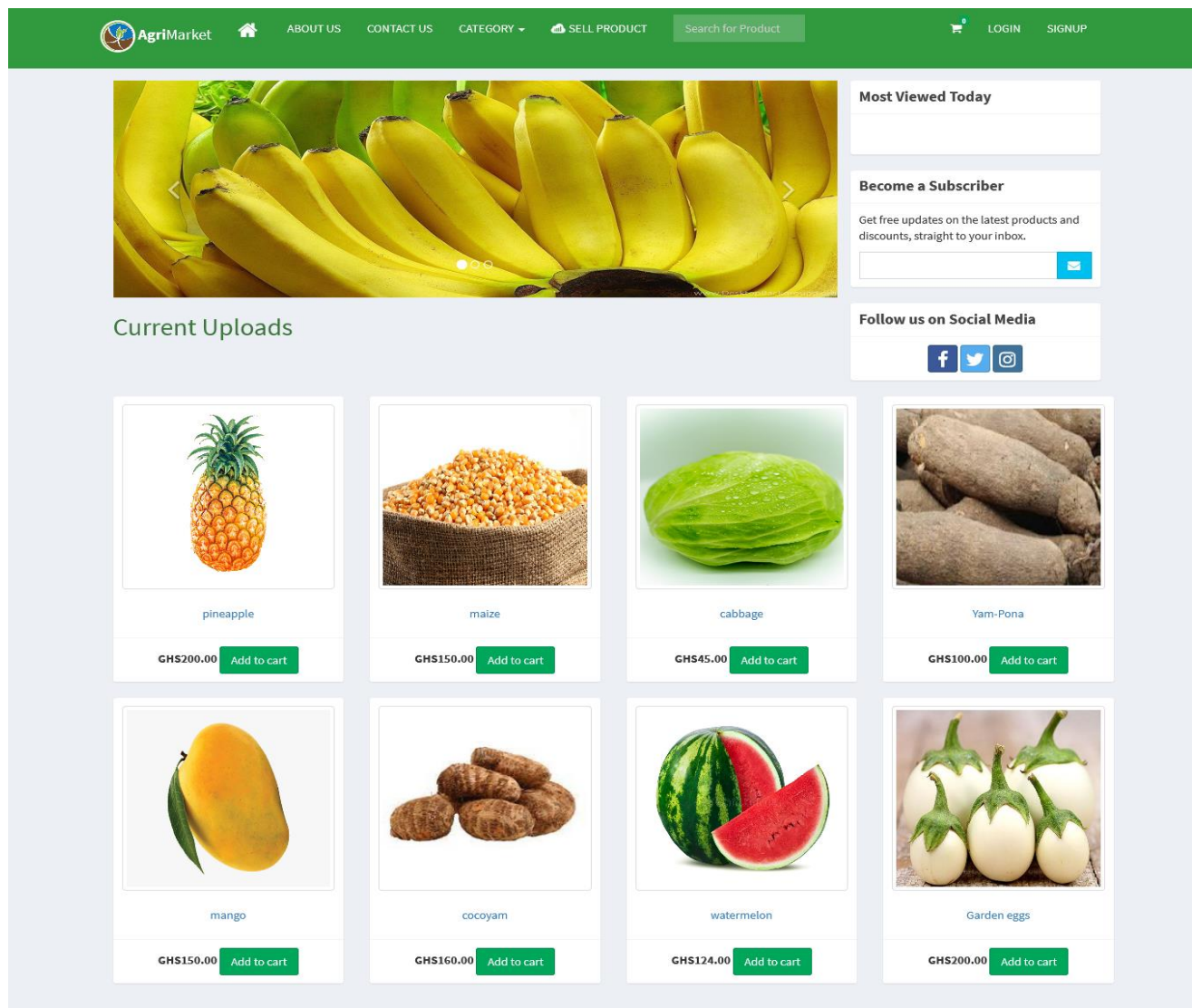
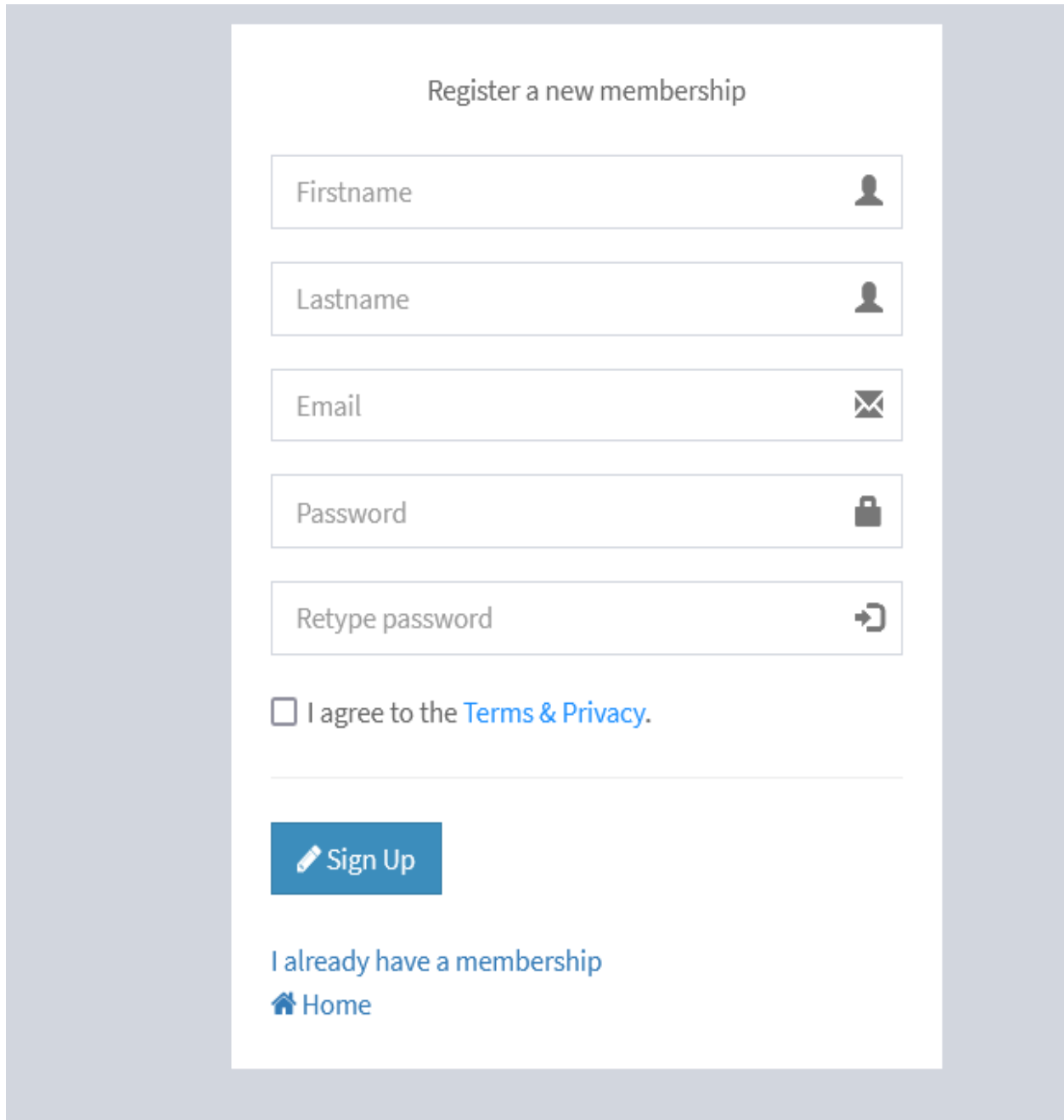


Figure 4. 13: Main page of the website

Signup Interface

The signup page is the form that gives the user access to create an account with the online agriculture marketing platform. It displays fields to accept user details, such as first name, last name, email etc.

The image shows a user registration form titled "Register a new membership". It contains five input fields: "Firstname" with a person icon, "Lastname" with a person icon, "Email" with an envelope icon, "Password" with a lock icon, and "Retype password" with a refresh icon. Below the fields is a checkbox labeled "I agree to the Terms & Privacy." with a horizontal line underneath. At the bottom is a blue "Sign Up" button with a pencil icon. Below the button are two links: "I already have a membership" and "Home" with a house icon.

Register a new membership

Firstname 

Lastname 

Email 

Password 

Retype password 

☐ I agree to the [Terms & Privacy.](#)

 Sign Up

[I already have a membership](#)

 [Home](#)

Figure 4. 14: User Signup page

Product Upload Page

It is the page a user visits to sell his product on the website. The user must have an account with the platform before he can upload his products to the site. However, his uploads are subject to the admin's approval.

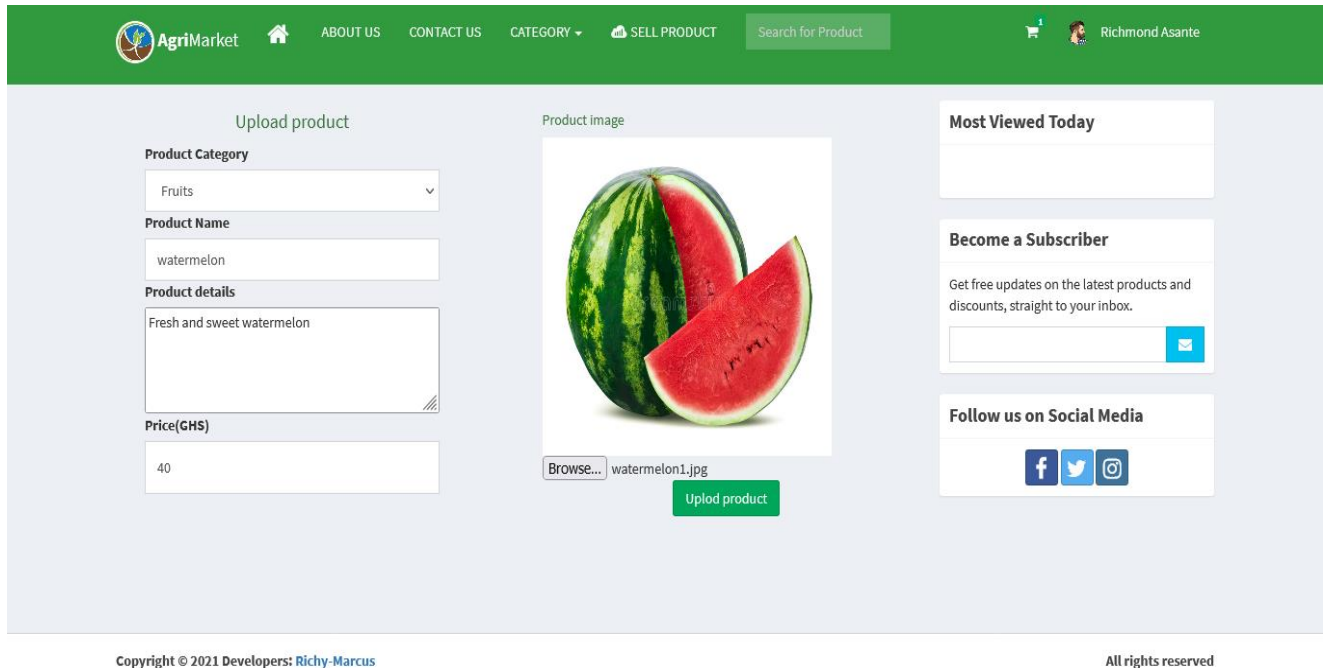


Figure 4. 15: Product upload page

Product Approval Page

The admin views the details of the uploaded product on this page to check the standard of the product, to approve or reject it. The details of the particular user who uploaded this product are displayed as well on this page.

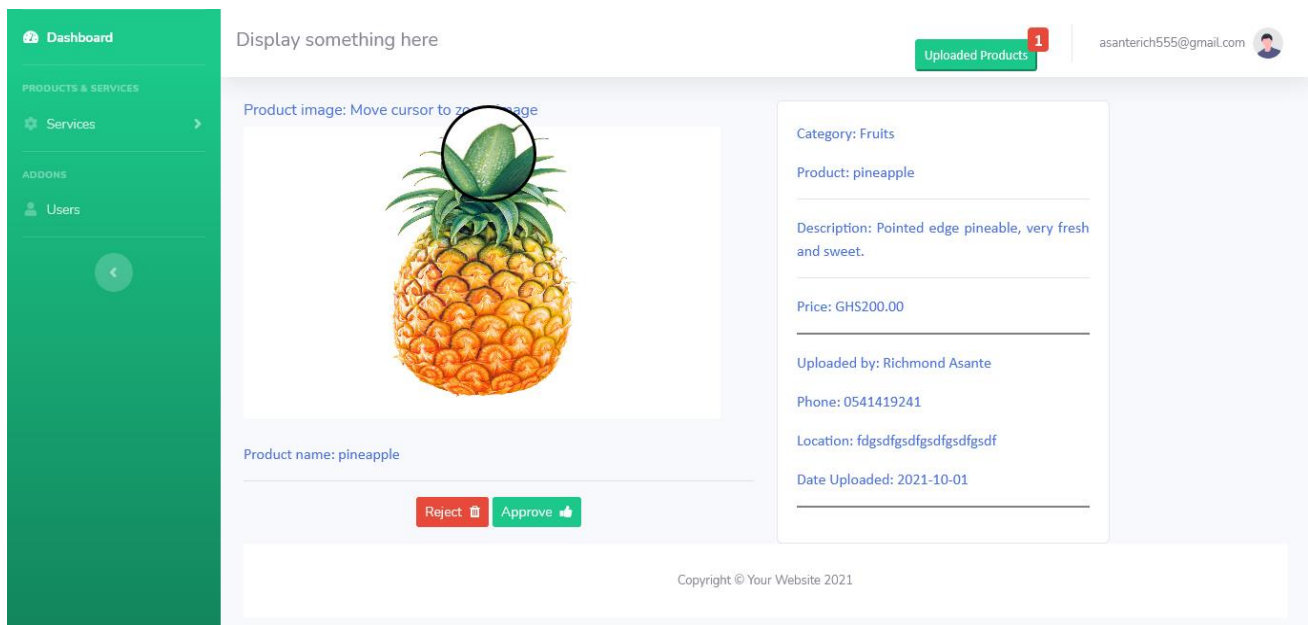


Figure 4. 16: Product approval page

Customer Cart Interface

The customer cart contains the products the customer has added to his wish-list to buy or review for later purchase. It displays a table showing the customer added products with their respective images, names, prices, and quantities is editable anytime. It also includes a button for making online payments for the products purchased.

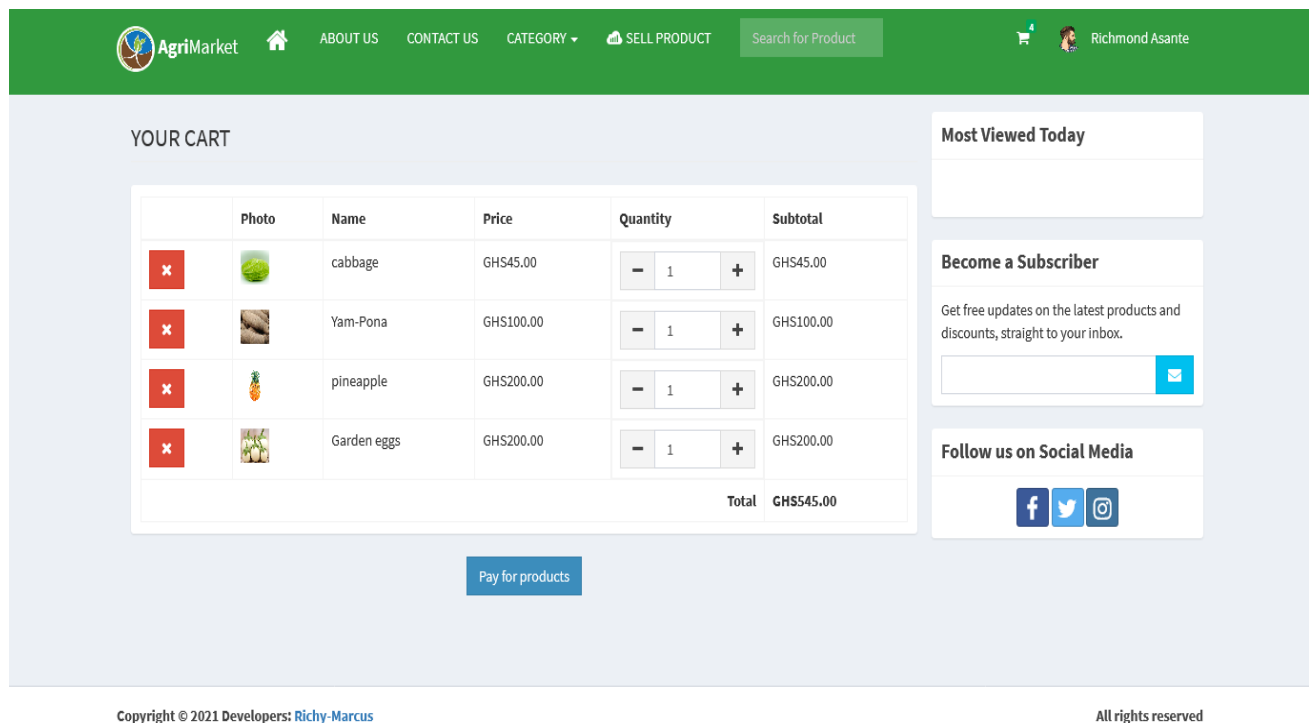


Figure 4. 17: Customer Cart Interface

Admin Main Page

It is the first page that when the admin logs into his account. It displays a table with transactions that have taken place, a button for adding product categories, and a button for viewing transactions summaries.

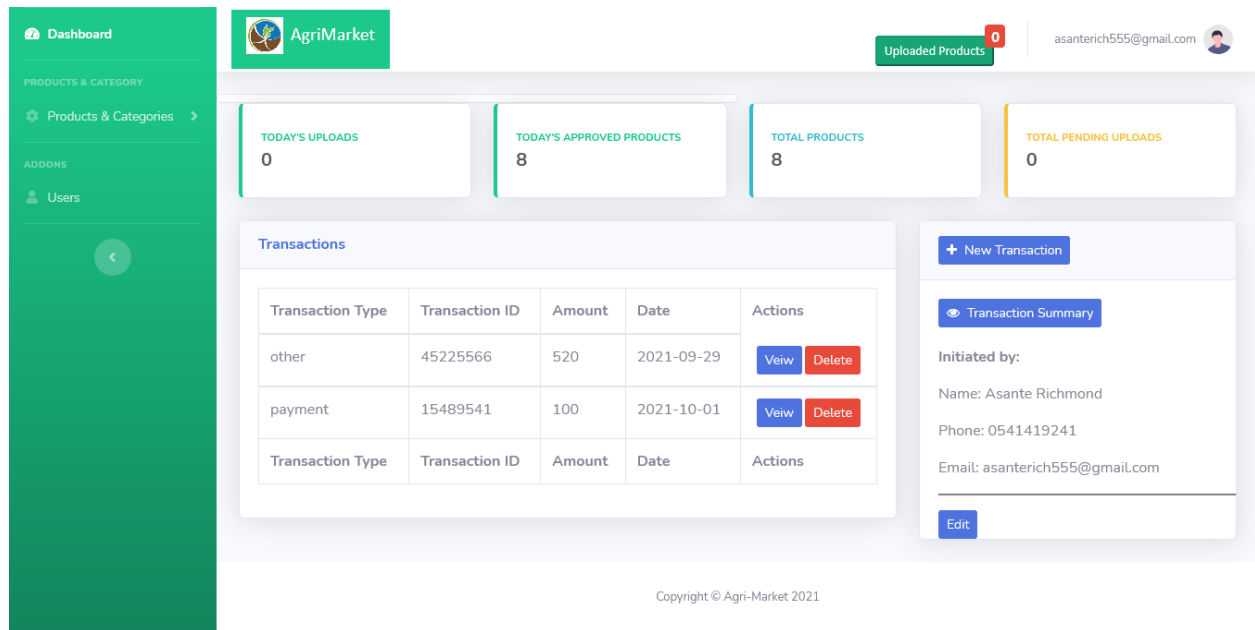


Figure 4. 18: Admin Main Page

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.0 Summary

This chapter contains the summary, conclusion and recommendations of the project. The existing system was analyzed, after which data were collected, followed by the analysis of data. Based on the analysis made limitations of the existing system were identified. It was followed by the system specification, system design, database design, user-side design, server-side design, interface design and system security design. The implementation of these design units was accomplished with the help of software development tools like Microsoft Visio, Sublime Text IDE, MySQL and Apache server. Php was used as the programming language and JavaScript as the scripting language. The front end was designed using HTML, CSS and Bootstrap.

After the design and implementation of the various units, system testing was carried out on each component and the general system using test data to check the expected and actual results. Finally, the performance evaluation of the system is certified since it is doing what it was designed to do.

5.1 Recommendation

We strongly recommend that users upload the actual images of their products coupled with precise details and reasonable product prices to ensure high customer trust and satisfaction, to guarantee their continual use of this system.

5.2 Future Works

The Agriculture Marketing System would be enhanced in future by adding a global positioning system (GPS); to certainly locate farmers and buyers.

5.3 Conclusion

This project aims to provide an online marketing system to be used by farmers and buyers of agricultural products within BAB Municipality. The aims include automating marketing, purchasing and payment of agriculture products. This project has been completed and will go a long way to change the mode of marketing and buying products of smallholder farmers in the area.

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APPENDICES

Appendix I

Signup Source Codes

```
<?php include 'includes/session.php'; ?>

<?php

if(isset($_SESSION['usermail'])){

    session_unset();

    session_destroy();

}

if(isset($_SESSION['user'])){

    session_unset();

    session_destroy();

}

?>

<?php include 'includes/header.php';

//require_once 'login.php';

?>

<body class="hold-transition login-page">

<div class="login-box">

    <?php

    if(isset($_SESSION['error'])){

        echo "

        <div class='callout callout-danger text-center'>
```

```

        <p>"$_SESSION['error']."</p>

    </div>

    ";

    unset($_SESSION['error']);

}

if(isset($_SESSION['success'])){

    echo "

    <div class='callout callout-success text-center'>

        <p>"$_SESSION['success']."</p>

    </div>

    ";

    unset($_SESSION['success']);

}

?>

<div class="login-box-body">

<?php

if(isset($_SESSION['message'])): ?>

    <div class="alert alert-<?=$_SESSION['msg_type']?>">

        <?php

            echo $_SESSION['message'];

            unset($_SESSION['message']);

        ?>

    </div>

```

```

<?php endif;

?>

<p class="login-box-msg">Sign in to start your session</p>

<form action="verify.php" method="POST">

    <div class="form-group has-feedback">

        <input type="email" class="form-control" name="email" placeholder="Email" re-
quired>

        <span class="glyphicon glyphicon-envelope form-control-feedback"></span>

    </div>

    <div class="form-group has-feedback">

        <input type="password" class="form-control" name="password" placeholder="Password"
required>

        <span class="glyphicon glyphicon-lock form-control-feedback"></span>

    </div>

    <div class="row">

        <div class="col-xs-4">

            <button type="submit" class="btn btn-primary btn-block btn-flat"
name="login">

                <i class="fa fa-sign-in"></i> Sign In</button>

        </div>

    </div>

</form>

```

```

<br>

<a href="password_forgot.php">I forgot my password</a><br>

<a href="signup.php" class="text-center">Register a new membership</a><br>

<a href="index.php"><i class="fa fa-home"></i> Home</a>

</div>

```

```

</div>

```

```

<?php include 'includes/scripts.php' ?>

```

```

</body>

```

```

</html>

```

Appendix II

Sample of Interviews Conducted

INTERVIEWER	RESPONDER
What Kind of crops do you grow?	I grow maize, tubers, plantain and vegetables
How do you sell your products in the market?	I just arrange the products and sit by them to wait for a buyer
Will it be beneficial to you to get an online platform to showcase your products to buyers from all walks of life?	Yes. It will encourage me to produce more
Have you sold or purchased any products online?	Yes. Mostly clothes, and electronic appliances at times

Table 5. 1: Interview with some of the farmers and buyers within the local market